

Taxes and Transfers

Economic Impact Analysis of Government

Edoardo Cefala



- Gender in Public Economics
- Taxes and Gender: Pink Tax, VAT, Income Taxation
- Labor Market: Gender Wage Gap & Child Penalties
- Taxes and Transfers in the Labor Market Model

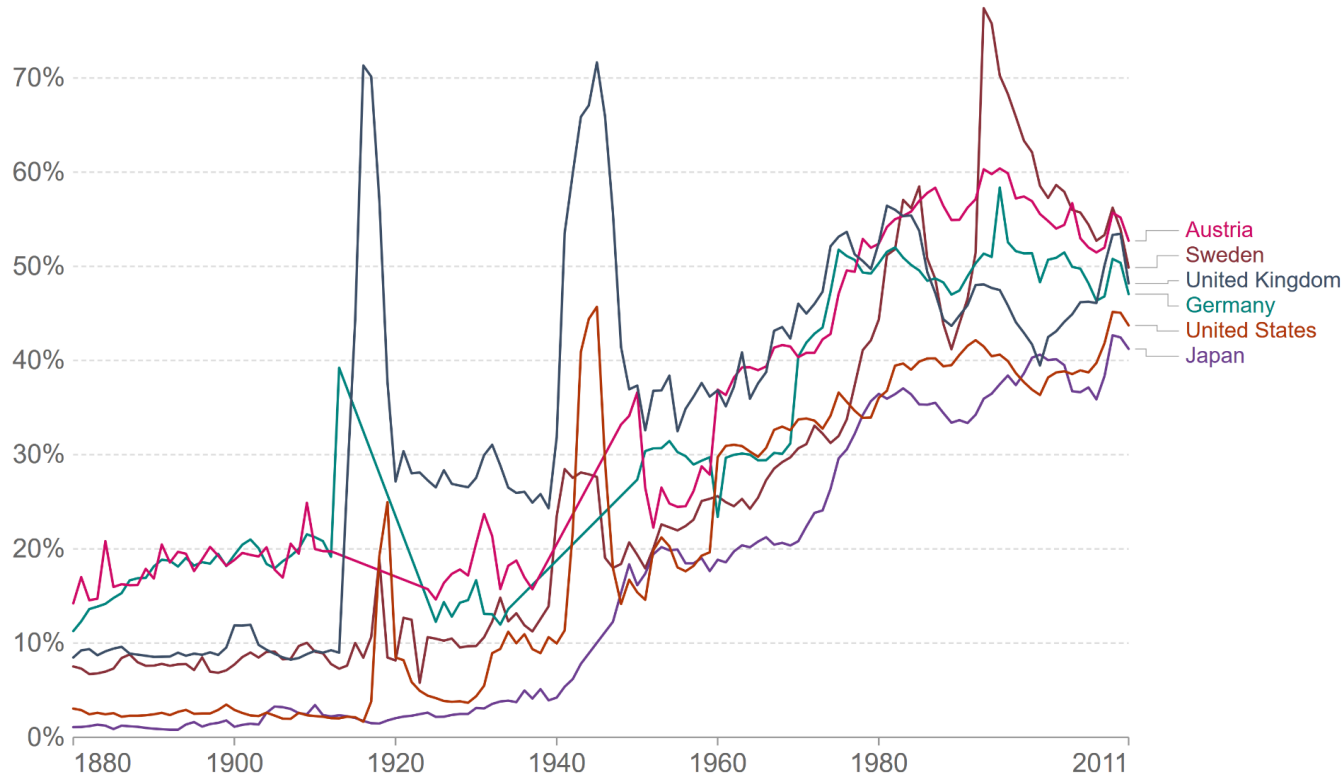
- Expansion of the welfare state and public provision after WWII
- Investments in childcare, education, care, healthcare
- significantly reduced (unpaid) workload for women

- Europe: since the 90s, an increased dismantling of the welfare state → responsibility has been pushed away from the state and back into the private sphere
- Examples: Discussion on regressive unemployment benefits in Austria, lower social benefits for part-time employees, Hartz IV in Germany, austerity especially in Southern Europe,...
- This dismantling affects people differently! e.g., care work during the pandemic was predominantly performed by women

Government Spending By country and time

Government spending, 1880 to 2011

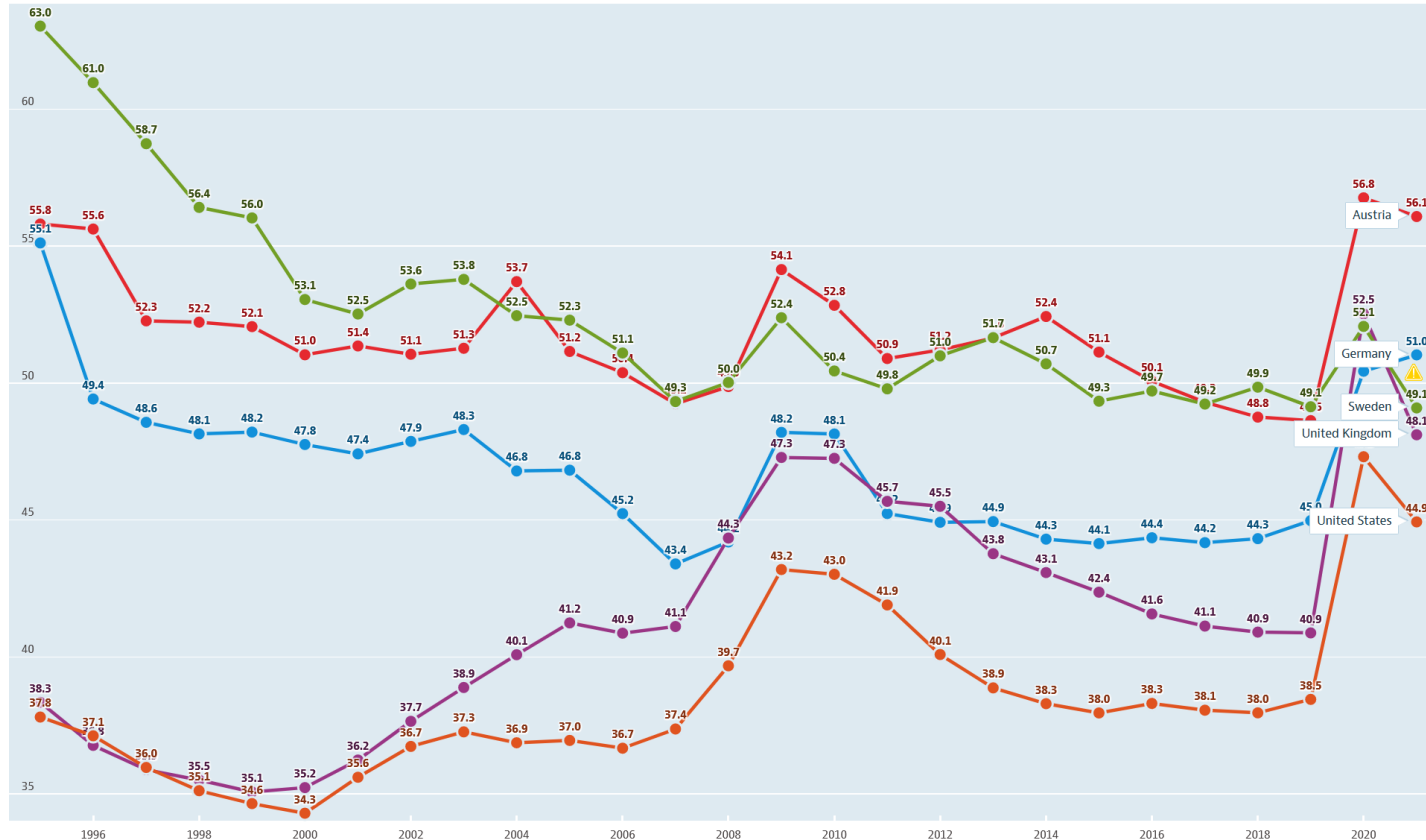
Total government spending, including interest government expenditures, as share of national GDP



- <https://ourworldindata.org/government-spending>
- <https://data.oecd.org/gga/general-government-spending.htm#indicator-chart>
- <https://www.imf.org/external/datamapper/exp@FPP/USA/FRA/JPN/GBR/SWE/ESP/ITA/ZAF/IND/TUR>

Government Spending By country and time

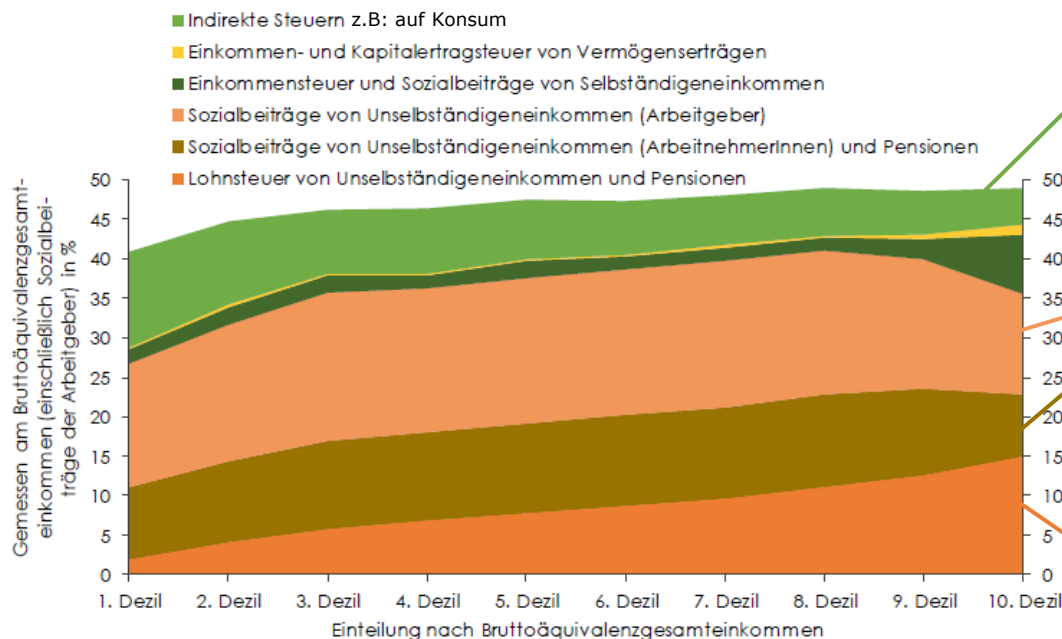
General government spending; Total, % of GDP, 1995 – 2021



OECD (2021), General government spending (indicator). <https://data.oecd.org/gga/general-government-spending.htm#indicator-chart>
(Accessed on 05 March 2024)

Redistributive Impact of the Public Sector in Austria I

Abbildung 14: Erwerbstätigenhaushalte – Äquivalente Abgaben einschließlich Sozialbeiträge der Arbeitgeber gemessen am Bruttoäquivalenzgesamteinkommen, 2015



Indirect Taxes:

- decreasing tax burden across all deciles (regressive)
- Reason: decreasing share of consumption expenditures with rising income

Social Security Contributions (SV-Beiträge):

- constant tax burden in the lower 9 deciles (proportional)
- decreasing tax burden in the last decile (regressive)
- Reason: capping of the contribution base (2022 at 5,670)

Wage Tax:

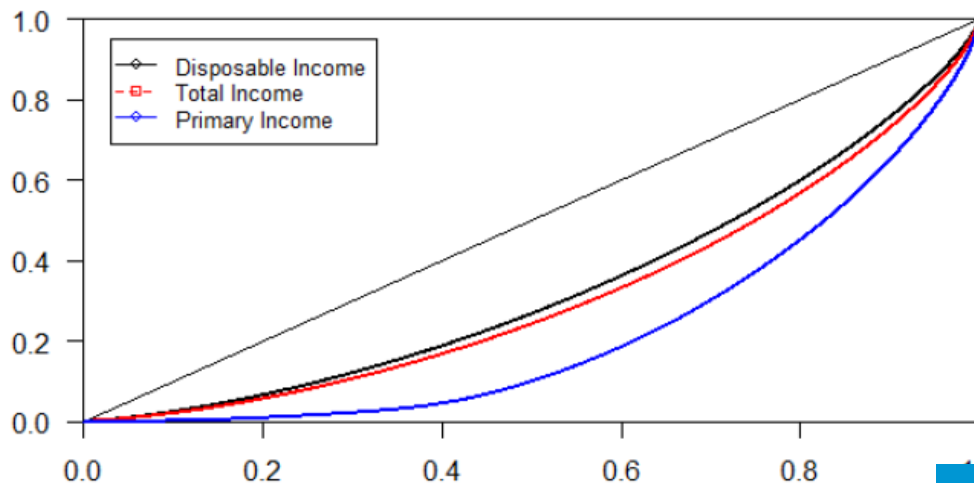
- increasing tax burden across all deciles (progressive)

Q: Statistik Austria, EU-SILC 2016, Konsumerhebung 2014/15; HFCS 2014; WIFO-Berechnungen.

Quelle: WIFO 2019, Umverteilung durch den Staat in Österreich 2015: S.57

Redistributive Impact of the Public Sector in Austria II

Lorenz Curves for different income aggregates [Source: EU-SILC]



- **Primary Income** = Market Income
- **Total Income** = Market Income **plus** received transfers
- **Disposable Income** = Net Income = Market Income **plus** received transfers **minus** paid levies

● → ●: Transfers are redistributive
● → ●: Levies are hardly redistributive

According to WIFO (2019):

Gini Market income: 0,38

Gini secondary income: 0,25

→ The lower the Gini, the more "equal" the income distribution

Quelle: EU-SILC 2012, eigene Darstellung

- The state's revenues and expenditures determine public activities
- = a regulatory instrument (Taxes!)
- Connects the state and the economy and shows the priorities & focus of current politics ≠ gender-neutral
- **Budget for Women and Equality**
- Currently: Federal Ministry for Women, Family, Integration and Media in the Federal Chancellery
- 2023: 24.3 million € (+ 5.9 million € or 32% compared to 2022)
- Focus: cross-departmental protection against violence
- Women's organizations criticize the too-low funding
- But: Many political measures have effects on women, not just women's and family policy (often not considered). Therefore → Gender Budgeting!

- "a gender-based assessment of budgets, incorporating a gender perspective at all levels of the budgetary process and restructuring revenues and expenditures in order to promote gender equality" (OECD 2016)
- **Review of public finances (budget process, revenues, and expenditures) for gender-specific effects → based on this analysis: restructuring of the public budget to achieve equality**
- **Goal: equitable distribution of financial resources with regard to gender**
- **Since May 2009, Gender Budgeting has been enshrined in the Austrian Federal Constitution (for the federal government, states, and municipalities)**
 - All ministries must consider equality goals in their budgets
 - Pilot projects in Austria: Meidling, Tulln,...
- **Since 2013, the goal of "actual equality of women and men" as part of the performance-oriented budget management of the federal government (Article 51, Paragraph 8 B-VG)**

Österreich Infos: <https://www.parlament.gv.at/PAKT/BUDG/PRAESENTATIONEN/index.shtml>

Österreich, Deutschland siehe: <https://awblog.at/geschlechtergerechtigkeit-steuern-ein-blick-auf-deutschland-und-oesterreich/>

OECD: <https://www.oecd-ilibrary.org/docserver/budget-16-5jfq80dq1zbn.pdf?expires=1617351285&id=id&accname=ocid177428&checksum=407FA2769DA49F69EF6B5A8E8F0FA5BD>

Gender Effects of Taxes



- Taxes can lead to discrimination based on gender
- **Consumption Taxation**
- Pink Tax
- Taxation of menstrual products
- **Income Taxation**
- Income splitting for married couples



Taxes and Gender

Consumption Taxation

- Pink Tax
- not a tax in the traditional sense, but a price markup on products/services that are marketed to people who are perceived as female
- Affected products include hygiene products, hairdressers, dry cleaning, toys
- Hardly any economic research on this topic, or no clear results



Taxes on Menstrual Products

Weltweit Maßnahmen to lower or abolish VAT

insufficient consumption of menstrual products can lead to health consequences, less participation in society, education, work, etc.

27.11.2020

EN

Official Journal of the European Union

C 411/38

P8_TA(2019)0014

Gender equality and taxation policies in the EU

European Parliament resolution of 15 January 2019 on gender equality and taxation policies in the EU (2018/2095(INI))

(2020/C 411/06)

21. Considers period poverty to be an ongoing issue in the EU, with Plan International UK estimating that 1 in 10 girls cannot afford sanitary products; regrets that female hygienic products, and care products and services for children, elderly people or people with disabilities, are still not considered as basic goods in all Member States; **calls on all Member States to eliminate the so-called 'care and tampon tax'** by making use of the flexibility introduced in the VAT Directive and applying exemptions or 0 % VAT rates to these essential basic goods; recognises that a reduction in price due to an exemption of VAT on these products would have an immeasurable benefit for young women; supports the movements undertaken to promote widespread sanitary supply availability and encourages Member States to **provide complementary feminine hygiene supplies in certain (public) spaces** such as schools, universities and homeless shelters, and for women from low-income backgrounds, with the aim of eradicating period poverty completely across EU public bathrooms;

Taxes and Gender

Consumption Taxation

- Case study Germany
- Tax reduction on sanitary pads & tampons from January 2020 (not on panty liners)
- Reduction from 19% (standard VAT rate) to 7% (reduced VAT rate)
- Tax reduction fully passed on, first pass-through already from the announcement

Taxes and Gender

Consumption Taxation

Frey and Haucap (2024) : „VAT pass-through: the case of a large and permanent reduction in the market for menstrual hygiene products“. *International Tax and Public Finance*

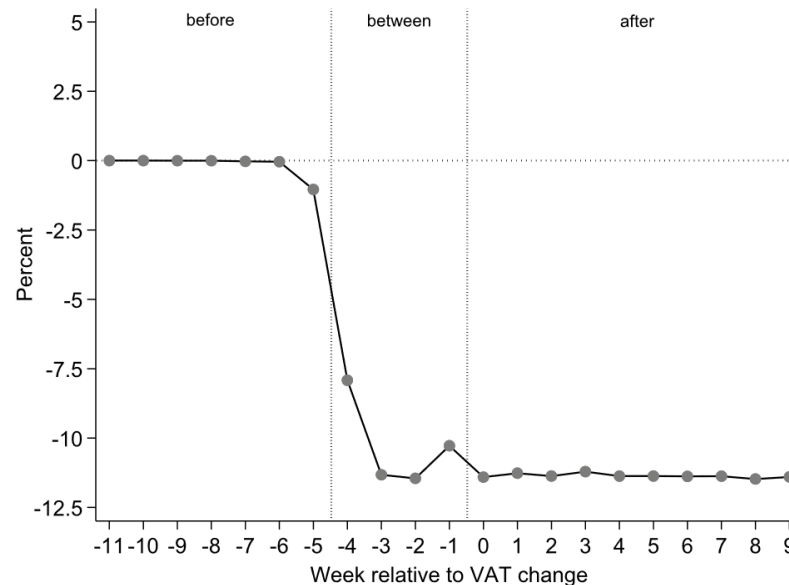


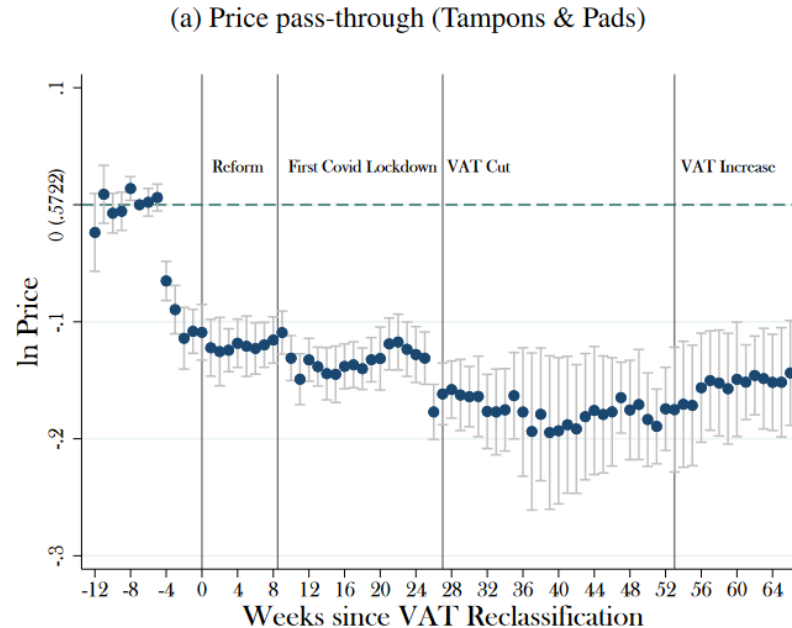
Fig. 1 This graph shows the price changes of menstrual hygiene products over time. Price changes are relative to $t = -11$. $t = 0$ represents the first week of 2020 when the law came into effect. $t = -5$ marks week number 48 in 2019 when the law was passed

Taxes and Gender

Consumption Taxation

Buettner et al. (2023): „Taxes and Gender Equality: The Incidence of the 'Tampon Tax'". *Cesifo Working Paper*

Figure 1: Tampons & Pads: Price and Unit-sales Dynamics around the Tax Reclassification



Taxes and Gender

Consumption Taxation

Kinnl and Wohak (2024): „Free the Period? Evaluating Tampon Tax Reforms Using Household Scanner Data”. *WU Working Paper*



Notes: Figure 3 presents estimation results from Equation 3 following Callaway and Sant'Anna (2021). It is based on household scanner data from 14 European countries. The Figure shows average price differences between each reform and non-reform countries in months relative to the announcement of the reform. The vertical lines indicate the month of the announcement (dashed line) and implementation (solid line). The estimation sample includes the continuous period from 6 pre-announcement to 12 post-implementation months. We cluster standard errors on product and country level.

- Individually (Austria) or at the household level (Germany)
- **Income splitting for married couples at the household level**
- the total taxable income of both partners is halved, the income tax is calculated, and the tax liability is then doubled → the greater the difference in earnings, the greater the savings
- Promotes primary earners, male-breadwinner model
- Austria: The sole-earner's tax credit (Alleinverdienerabsetzbetrag) and Family Bonus+ promote an unequal division of paid work

Income Splitting for Married Couples

Zu versteuerndes Jahresbruttoeinkommen
eines Paares in allen Fällen 100 000 Euro



Jährliche Einkommensteuer
eines **unverheirateten** Paares

Fall 1

Mann verdient 100 000 €,
Frau verdient nichts

33 605 €

Fall 2

Mann verdient 70 000 €,
Frau verdient 30 000 €

26 473 €

Fall 3

Mann verdient 60 000 €,
Frau verdient 40 000 €

25 631 €

Fall 4

Mann verdient 50 000 €,
Frau verdient 50 000 €

25 272 €

Vorteil des Ehegattensplittings

8333 €

1201 €

359 €

0 €



Jährliche Einkommensteuer
eines **verheirateten** Paares

25 272 €

25 272 €

25 272 €

25 272 €

Quelle: Bund der Steuerzahler / F.A.Z.-Grafik Brouker

Individual vs. Household Taxation

Empirical Evidence

- Buettner et al. (2019). "Tax planning of married couples and intra-household income inequality". *Journal of Public Economics*
- How does the income distribution between spouses affect the choice of tax model?
- Data from wage and income tax statistics
- symmetric vs. (fe)male-favoring taxation of the incomes of spouses in Germany
- Differences in the choice of tax model when the primary earner is male or female

Individual vs. Household Taxation

Empirical Evidence

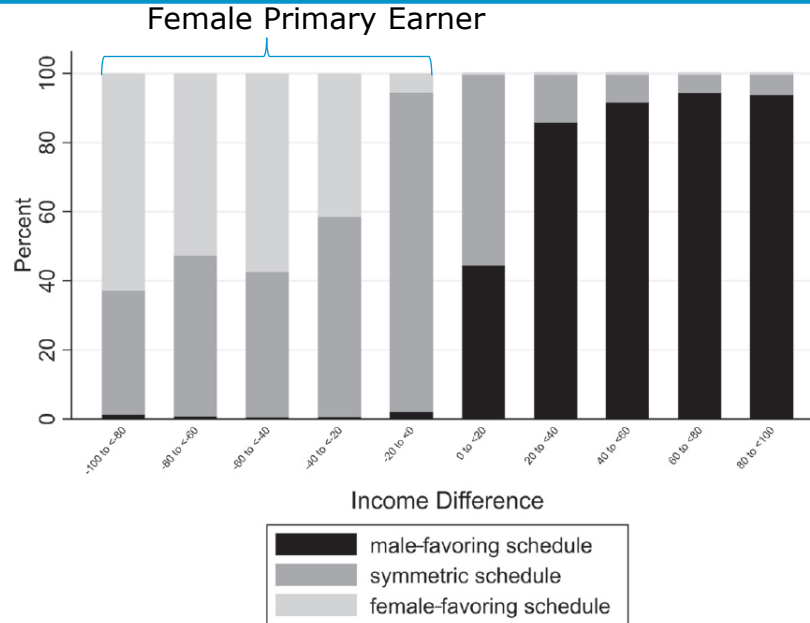


Fig. 1. Choice of Payroll-tax treatment and income difference between husband and wife. Share of all households in the dataset that have chosen a male-favoring, symmetric, or female-favoring payroll-tax treatment by labor income difference between husband and wife, weighted with survey weights. The horizontal axis reports the gross labor income difference in thousand euros. Statistics are based on a random sample drawn from the whole population of the income tax returns filed for the year 2004 in Germany. The number of observations varies between income classes. The following table reports the number of observations by the respective income difference between husband and wife:

Wife is primary earner					
Income difference (in 1,000 euros)	-100 to -80	-80 to -60	-60 to -40	-40 to -20	-20 to 0
Obs.	574	1,220	2,736	6,660	20,314
Husband is primary earner					
Income difference (in 1,000 euros)	0 to 20	20 to 40	40 to 60	60 to 80	80 to 100
Obs.	38,978	38,085	32,928	30,566	22,800

Due to data protection issues, for couples with income difference beyond 100,000 euros, the choice of the payroll-tax treatment was not available for intervals of the income difference.

- When a man/woman is the primary earner, the model that favors them is more likely to be chosen.
- But: Asymmetrical! If a woman is the primary earner, at least 35% of households are still in symmetrical taxation.
- Social norms?

Individual vs. Household Taxation

Empirical Evidence

- Tax avoidance as a key reason for the choice of the respective model
- Nevertheless: a large number of households do not choose a tax model in a way that minimizes the tax burden
- The loss of income of secondary earners is a driving factor for why the tax-minimizing model is not chosen. This is because secondary earners pass on their tax-free allowance to the primary earner.

Gender-Effekte am Arbeitsmarkt



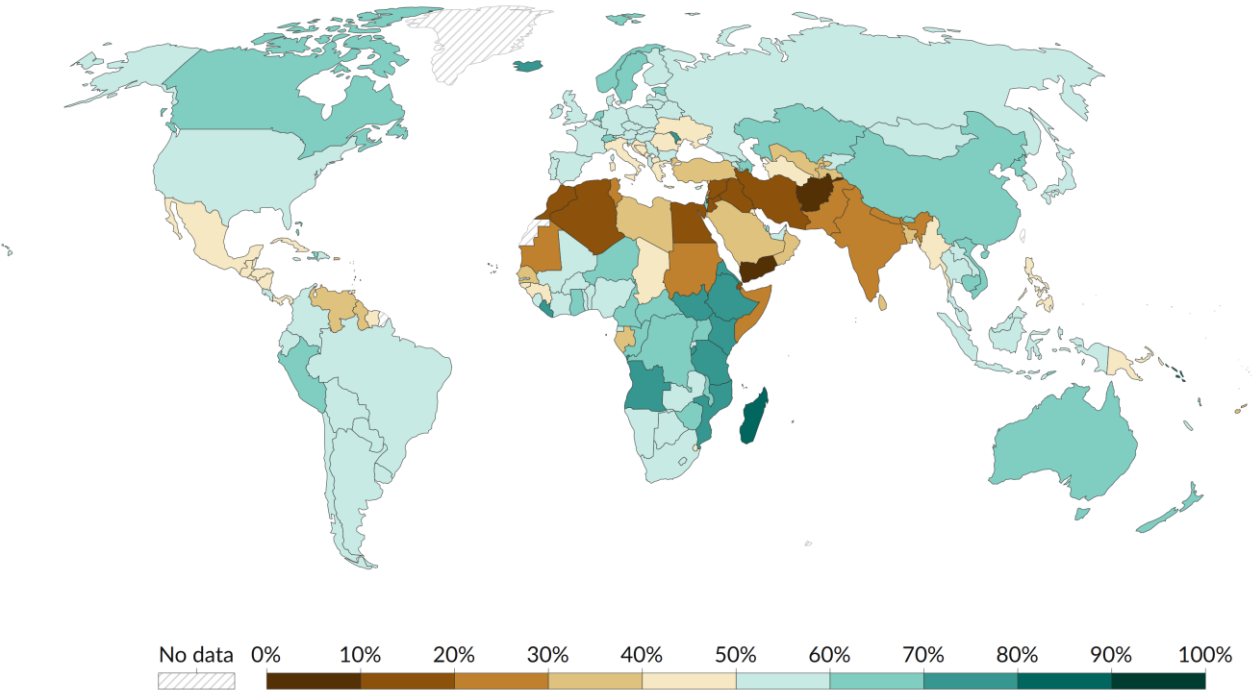
- **Labor Market**
- Women participate less often in the labor market
- Women work fewer hours → **Division of labor: paid employment, unpaid reproductive labor**
- Women earn less → **Income: Gender Wage Gap**
- Women are underrepresented at the top of the career ladder → **horizontal/vertical segregation, "sticky floor, Glass Ceiling"**
- **Capital, Assets:** Gender Wealth Gap
- **Poverty**
- Pension: Gender Gap in Pensions, old-age poverty
- Crises, austerity

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Female labor force participation rates, 2022

Labor force participation rate is the proportion of the population ages 15 and older that is economically active.

Our World
in Data



Data source: Multiple sources compiled by World Bank (2024)

Note: All figures correspond to 'modeled ILO estimates' (see source for details).

OurWorldinData.org/female-labor-supply | CC BY

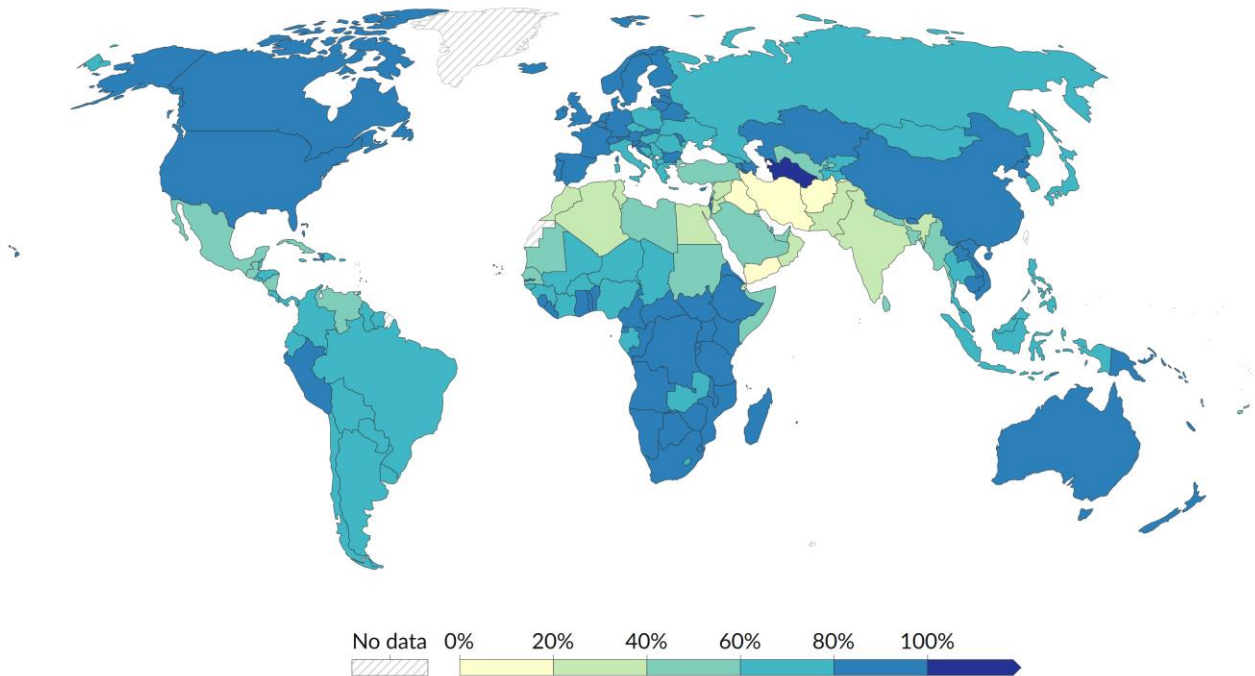
Labor Market

Paid Work

Ratio of female to male labor force participation rates, 2022

Our World
in Data

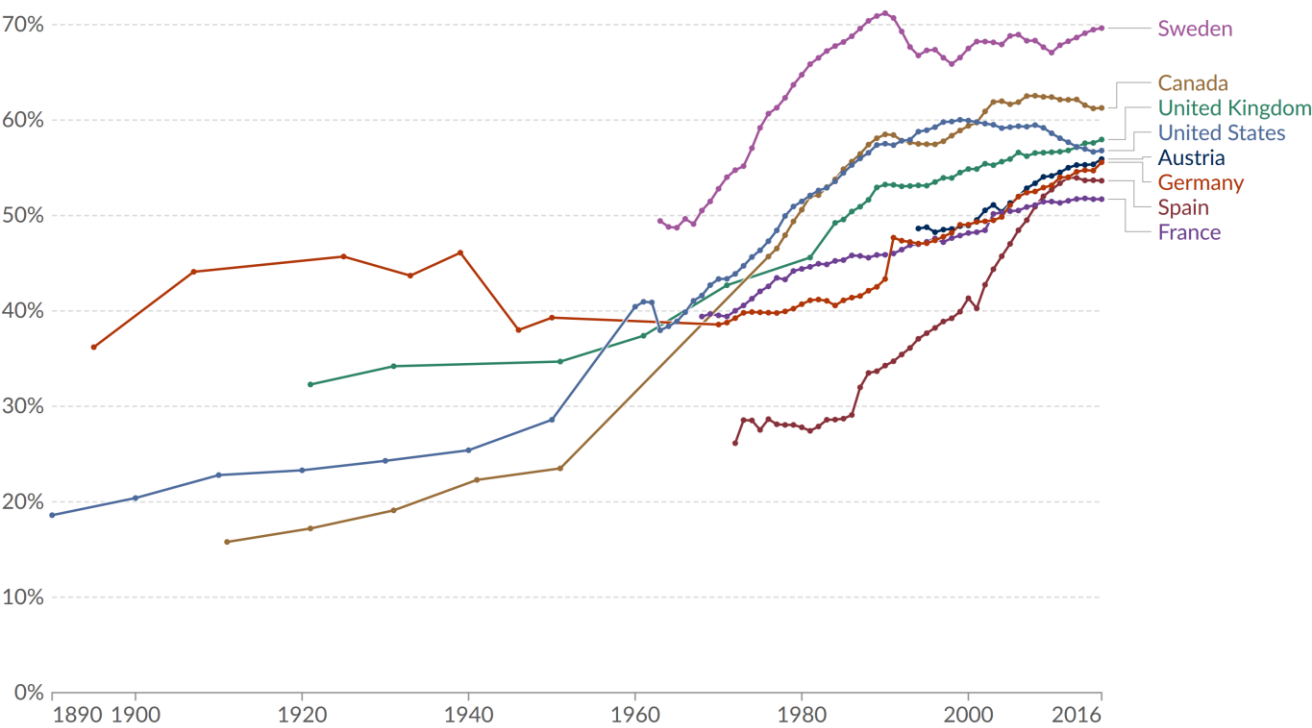
This ratio is calculated by dividing the labor force participation rate among women by the corresponding rate for men. The labor force participation rate is the proportion of the population aged 15 years and older that is economically active.



Share of women in the workforce

Proportion of the female population aged 15+ that is economically active.

Our World
in Data



Data source: Our World In Data based on OECD (2017) and Long (1958)

OurWorldinData.org/female-labor-supply | CC BY

Note: For some observations prior to 1960, the participation rate is calculated with respect to the female population aged 14+.

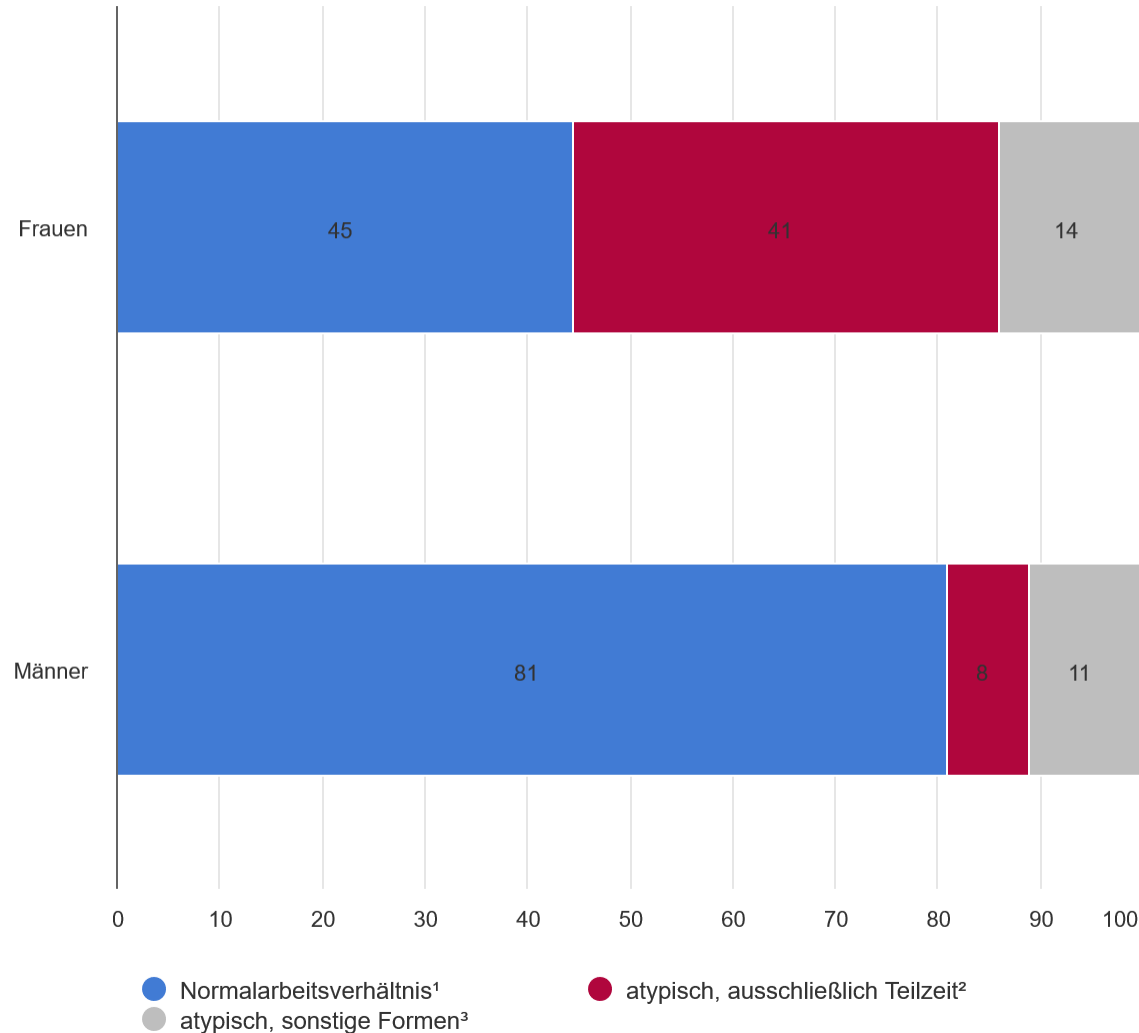
Why are we observing an increase here?

- Equality laws & measures
- Household technologies
- Structural changes
- Opportunities for family planning, contraception

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Labor Market Paid Work

- **Core Labor Market and Atypical Employment**
- Part-time, marginal employment, temp work, new self-employed, freelancers, and fixed-term contracts
- **Employment Rate 15-64 Yrs (Austria 2023)**
- Men: 77.9%
- Women: 70.3%
- **Part-time Rate (Austria 2023)**
- Men: 13.4%
- Women: 50.6%
- → Part-time work remains female, 77.1% of part-time employees are women
- → Impacts on social security claims

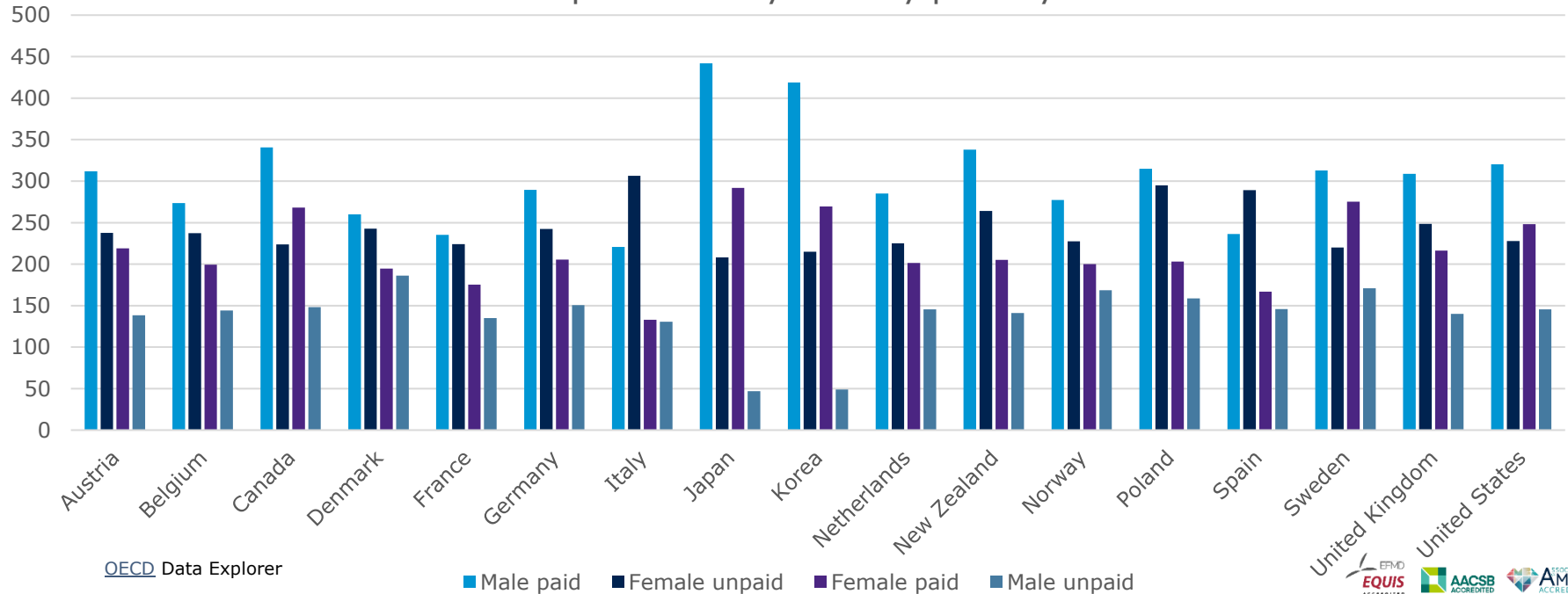


Labor Market

Unpaid Work

- Who does unpaid work?
- In most countries, for minutes worked per day, the following applies: men paid > women unpaid > women paid > men unpaid

Paid and Unpaid work by country per day



- What is unpaid work?
- **Third-Person Criterion:** unpaid work = as soon as the activity could be purchased from a third person on the free service market
- Enables a clear distinction from personal and leisure activities
- Unpaid work is not monetarily compensated, although a service is produced (not measured in GDP)
- Work: Work is understood as paid work/employment, which aims to generate income. In neoclassical economics, work is a key factor of production alongside capital, land, and knowledge.
- But: **without unpaid work, no paid work is possible!** The value creation in households accounts for an additional 30-50% of the existing GDP (A&Wblog 2019).

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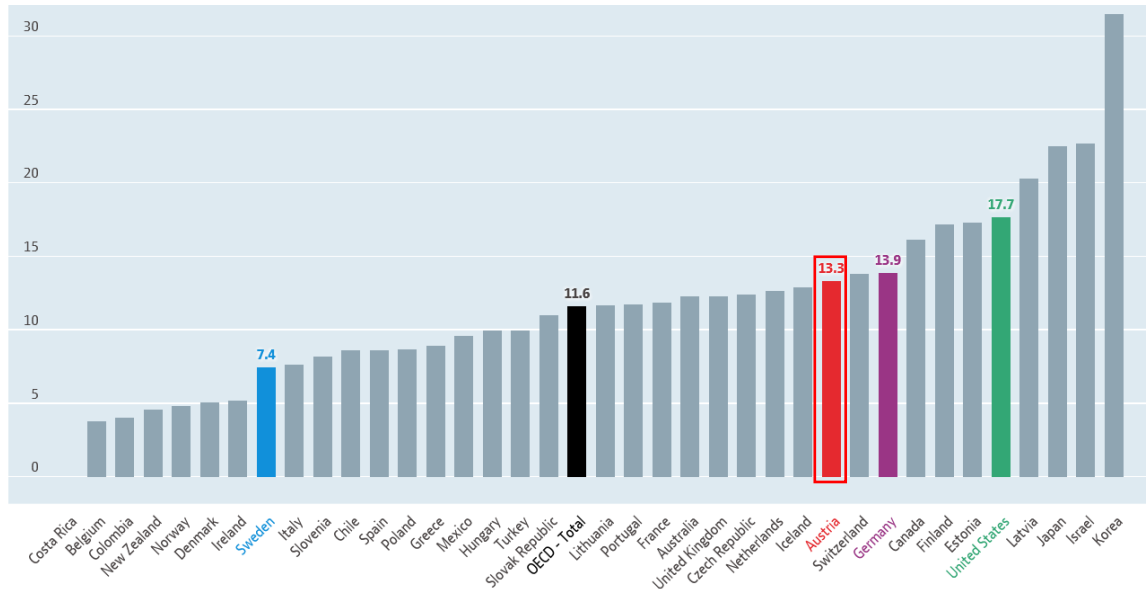
- **Popular measure, but what does it say?**
- Difference between median or average income of women and men, relative to income of men $\rightarrow \frac{Y_M - Y_F}{Y_M}$
- Unadjusted: compares gross hourly wages
- **What causes the Gender Wage Gap?**
- Industry, occupation, education, age, experience, working hours, career interruptions, discrimination, norms, lack of role models, networks, segregation, negotiation skills, wage (in)transparency... play a role
- Partially considered in the adjustment, but can only explain a part of the gap
- Endogeneity problem: e.g., is industry segregation already a result of structural inequalities?

Labor Market

Gender Wage Gap

Gender wage gap Employees, Percentage, 2020 or latest available

Source: Earnings: Gross earnings: decile ratios



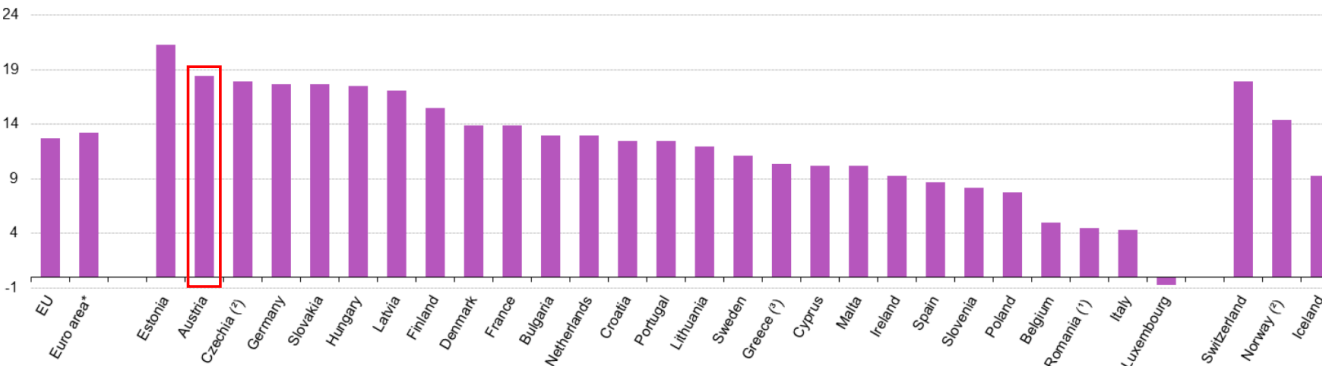
- Median wage is used for the calculation
- What happens if we use the average wage?

Labor Market

Gender Wage Gap

The unadjusted gender pay gap, 2022

(difference between average gross hourly earnings of male and female employees as % of male gross earnings)



Note: For all the countries except Czechia and Iceland: data for enterprises employing 10 or more employees, NACE Rev. 2 B to S (-0); Czechia: data for enterprises employing 1 or more employees, NACE Rev. 2 B to S; Iceland: NACE Rev. 2 sections C to H, J, K, P, Q. Gender pay gap data for 2022 are provisional until benchmark figures, taken from the Structure of Earnings survey, become available in December 2024.

* Euro area (2015-2022)

(*) Estimated data.

(*) Definition differs (see metadata).

(*) 2018 data.

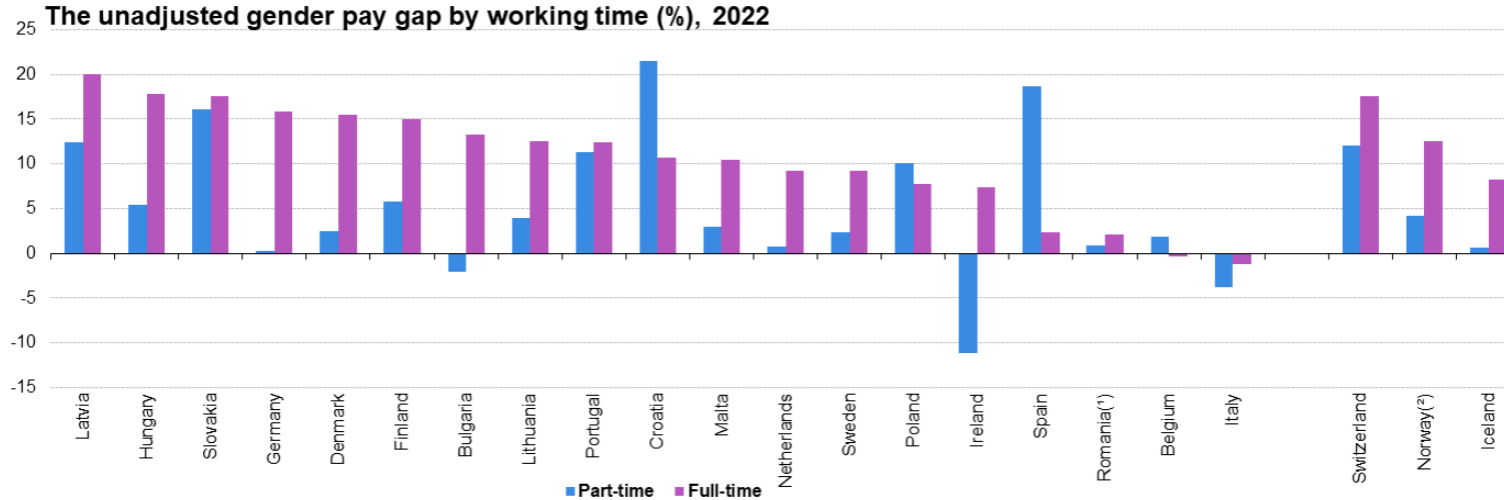
Source: Eurostat (online data code: sdg_05_20)

- If we take the average wage, the gap increases → high outliers (more often men) have a stronger weighting!
- Unadjusted Gap = explained + unexplained (adjusted) Gap
- Detailed calculations for the gap

here: <https://ec.europa.eu/eurostat/documents/3888793/13484385/KS-TC-21-004-EN-N.pdf/69965821-22ed-7c56-c859-cd7b10e011c5?t=1633341826751>

Labor Market

Gender Wage Gap



Note: For all the countries except Iceland: data for enterprises employing 10 or more employees; NACE Rev. 2 B to S (-0);

Iceland: NACE Rev. 2 sections C to H, J, K, P, Q. Data not available for CZ, EE, EL, FR, CY, LU, AT and SI.

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⁽¹⁾ Estimated data.

⁽²⁾ Definition differs (see metadata).

Source: Eurostat (online data code: earn_gr_gpgr2wt)

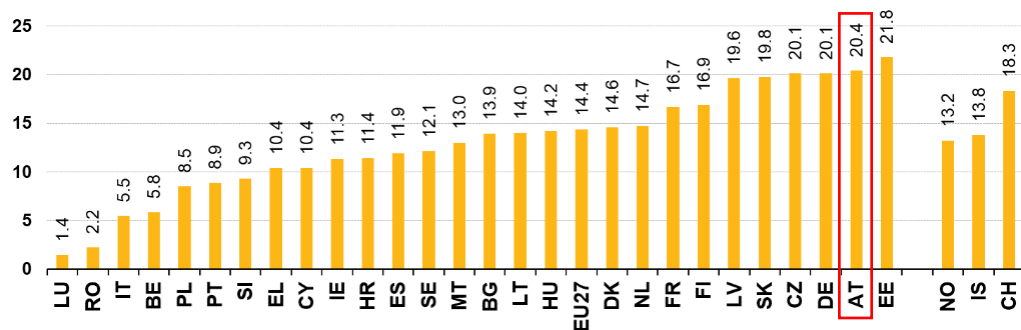
eurostat 

Women in part-time work have a comparatively higher qualification and education level than men in part-time work → different reasons for part-time work!

Labor Market

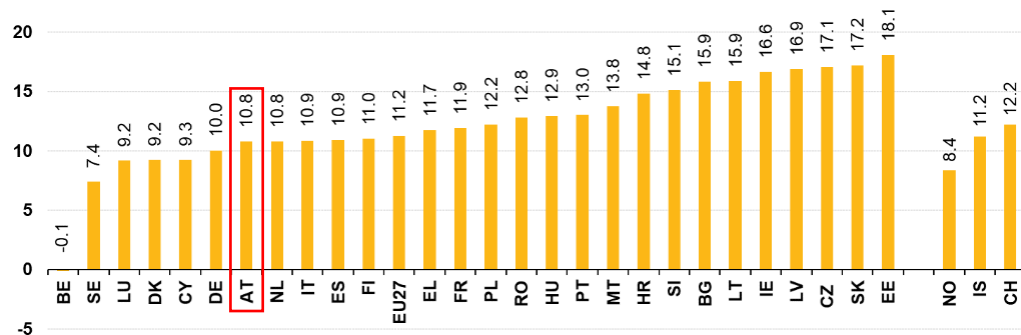
Gender Wage Gap

Figure 2a: Unadjusted gender pay gap (%), 2018



2018: approx. 50% of the Gender Wage Gap cannot be explained.

Figure 2b: Unexplained gender pay gap (%), 2018

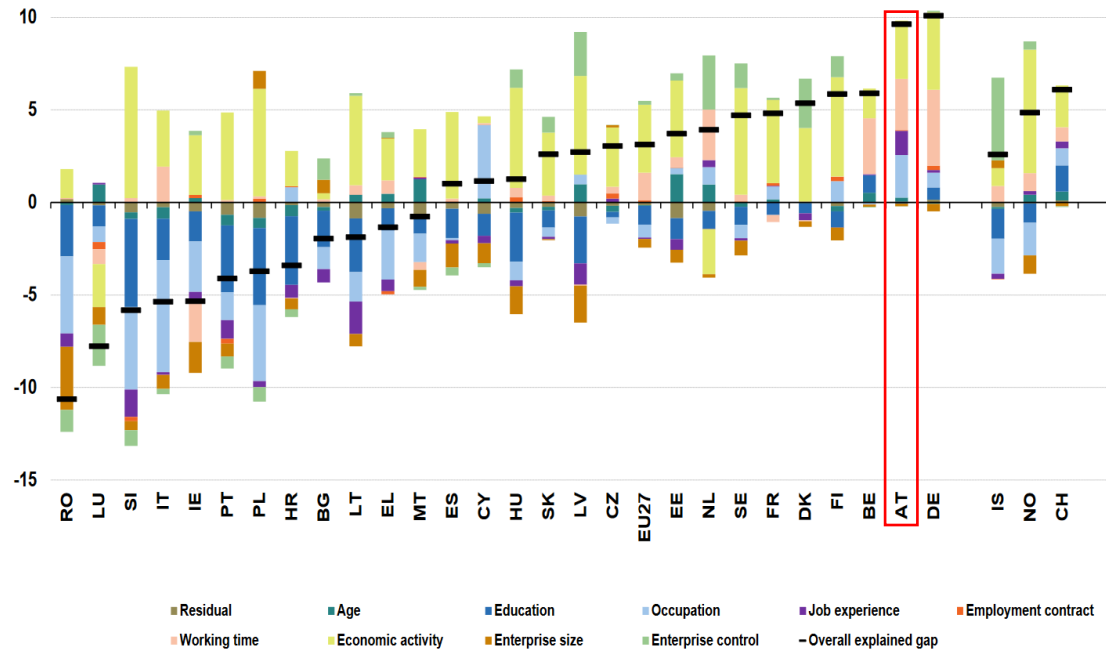


Unexplained/Adjusted = that part of the gap that cannot be explained by labor market characteristics

Labor Market

Gender Wage Gap

Figure 3: Decomposition of the explained gender pay gap, 2018



- **Positive Gap** largely explained by:
 - Economic activity (Sector usual pay) = Horizontal Segregation
 - Working time (Full- vs. Part-time)
- **Negative Gap** by:
 - Education
 - Occupation (not in Austria) = Vertical Segregation
 - → On average, women in the labor market are more highly educated → higher expected wages

Labor Market

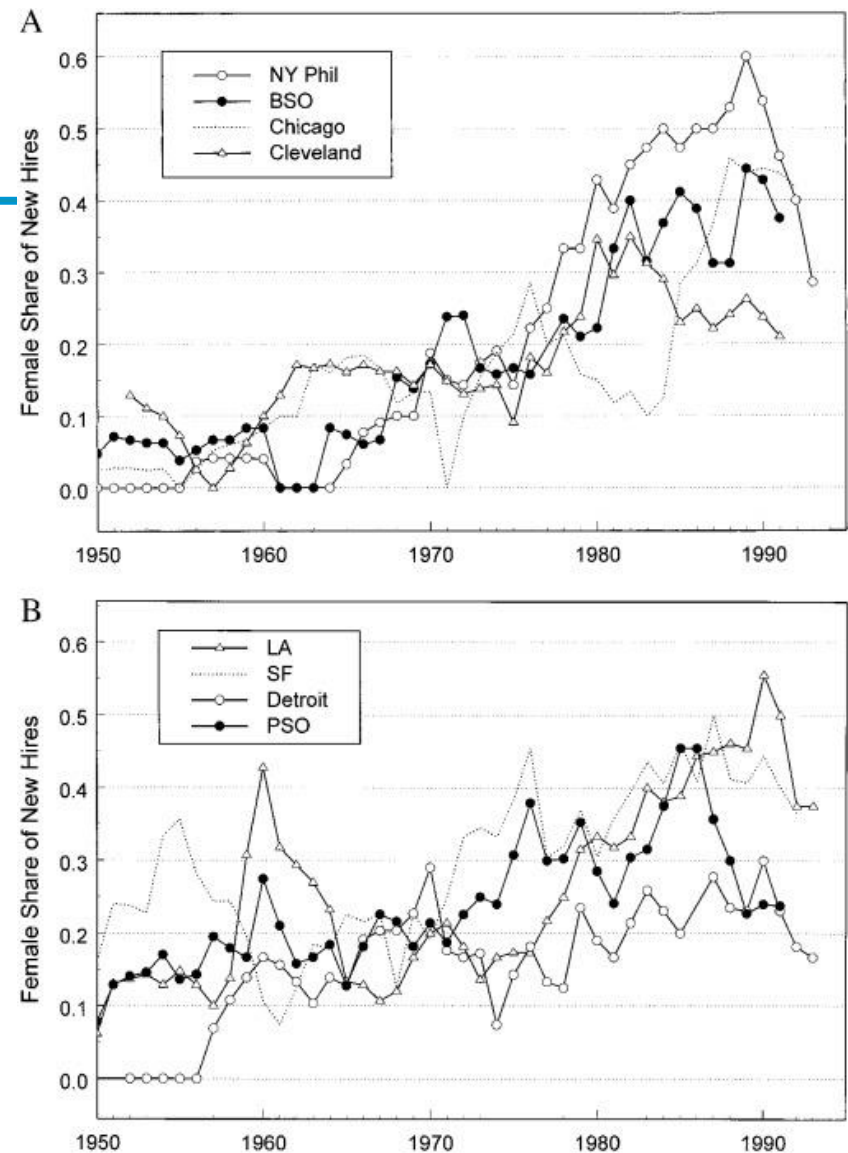
Gender Wage Gap

- Unexplained Gap does not measure pure discrimination!
- Missing information about qualifications (quality of training, effort, motivation)
 - If these factors are not (or cannot be) taken into account in the data, the "unexplained" part may also reflect these unobserved factors
 - → Over/Underestimation of discrimination
- Controlling for factors that are already a result of discrimination (industry, occupation).
 - If these factors are included as control variables, the analysis could "explain away" inequalities that are already a result of discriminatory structures
 - → Underestimation of discrimination

- How can we measure discrimination?
- Goldin & Rouse (2000): "Orchestrating Impartiality: The Impact of "Blind" Auditions on Female Musicians". *American Economic Review*
- Orchestra Auditions: Major orchestras in the USA changed their audition process in the 1970s and 1980s.
- Auditions with committees of multiple members

Labor Market Discrimination

- **Results:**
- Introduction of blind auditions -> increases the probability that women remain in the selection process and are hired by the orchestra.
- The change can explain about one-third of the increase in the share of women among new hires.
- We see: There was discrimination before the introduction of blind auditions.



Labor Market Discrimination

Petit (2007): "The effects of age and family constraints on gender hiring discrimination: A field experiment in the French financial sector". Labour Economics January-March 2002: Applications for low- and highly-qualified positions in the administrative or commercial sector -> Only difference in the documents: gender Comparison of access to job interviews for single and childless male and female applicants aged 25 and 37

Results

Women aged 25, who apply for highly qualified positions in administration, are discriminated against.

Young men are favored over young women for long-term contracts. No gender discrimination for older applicants aged 37, but fewer invitations. Deep ResearchCanvasImagineApprendimento guidato

Labor Market Discrimination

(Application A)

Caroline PETIT

Address: (75005 PARIS)

Phone number:

E-Mail:

(25, single, childless)

PROFESSIONAL EXPERIENCE

Since March 2000

BANK CUSTOMER ASSISTANT / BANK A, PARIS
Controlling and keeping track of a portfolio made up of
80 customers

Aug. 1999 to Feb. 2000

Banking and financial product sales and advice
COUNTER CLERK / BANK A, PARIS
(Fixed-term contract)

April 1999 to June 1999

Handling cash, cheques and change at the counter
Basic advice on banking services
RECEPTIONIST / BANK B, PARIS

(Application B)

Arnaud FONTAINE

Address : (75007 Paris)

Phone number :

E-Mail :

(24, single, childless)

PROFESSIONAL EXPERIENCE

Since Nov. 1999

BANK 1 – Paris

Bank customer assistant

Keeping track of a client portfolio (90 customers)
Explaining banking services and financial product
offers on the phone

Credit simulations proposals

Feb. 1999 to Oct. 1999

BANK 2 – Paris

Counter clerk

Current operations at the bank counter
Clients' reception

March 1997 to Jan. 1999

FIRM 3 (insurance company) – Paris

General clerk (part-time job)

- **Wechselbaumer (2004): "Is It Sex or Personality? The Impact of Sex Stereotypes on Discrimination in Applicant Selection". Eastern Economic Journal**
- Identical resumes, different gender and "gender-specific" attributes
- What role do gender and personality play?
- 3 groups: Male & male attributes, female & female attributes, female & male attributes
- **Results**
- In female-dominated occupations, women are more likely to receive a callback than men – regardless of their attributes
- In male-dominated occupations, women are less likely to receive a callback – independent of their attributes → Discrimination based on gender and not on attributes

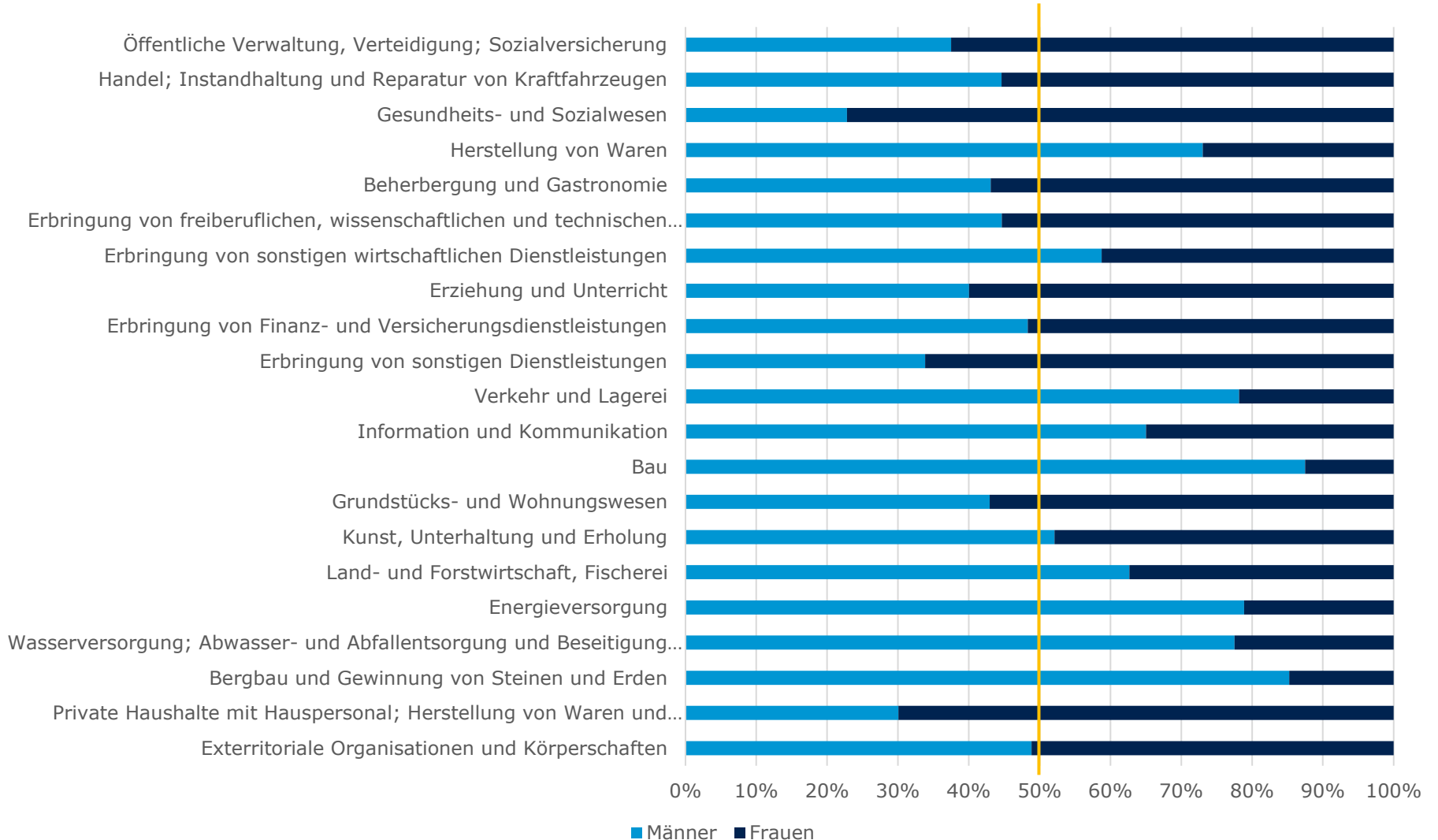
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Labor Market

Segregation in the Labor Market

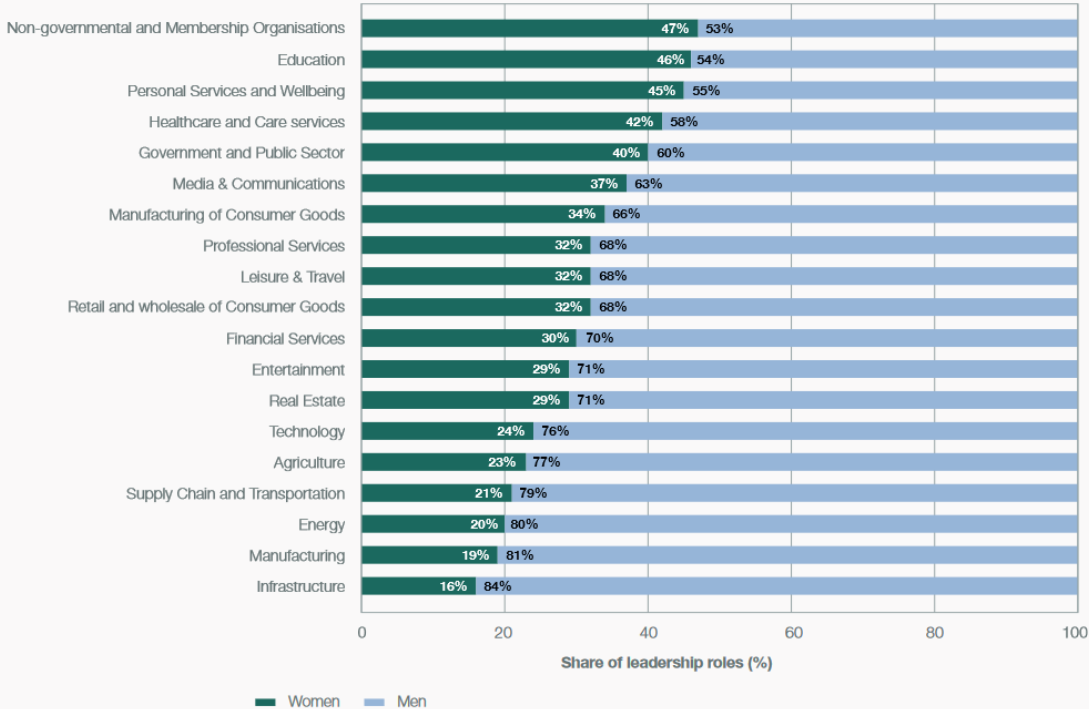
- **Horizontal Segregation**
- Typical "female and male industries"
- Industries in which more women are employed tend to be paid less
- **Vertical Segregation**
- Women & men pursue different activities within industries
- Men are much more often in leadership positions (even in "female industries")
- "Glass Ceiling"

Frauen- & Männeranteile nach Branchen



Labor Market Segregation in the Labor Market

FIGURE 2.7 | Women in leadership roles, by industry (%)
2022



Leadership
31% women on average
increasing over time

Women are more likely to be in leadership positions in sectors where they are already well-represented

Source
LinkedIn Economic Graph

Note
Leadership roles included in the sample are Director, VP, CXO, Partner. The sample includes data from 155 countries.

Labor Market Segregation in the Labor Market

How can horizontal segregation be counteracted?

Card et al. (2024): "Gender Preferences in Job Vacancies and Workplace Gender Diversity".
The Review of Economic Studies

- In 2005, the Equal Treatment Ombudsman's Office in Austria launched a campaign to point out that gender-specific preferences in job advertisements are illegal.
- 40% of the job advertisements on the country's largest job board listed a gender preference; within one year, the share dropped to under 5%.

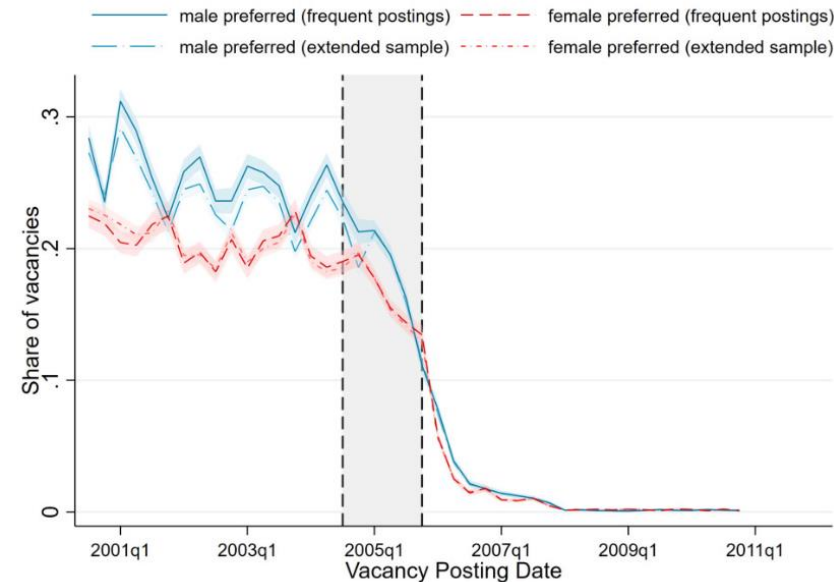


FIGURE I
Use of stated gender preferences in posted vacancies

Labor Market Segregation in the Labor Market

- Women are more likely to be hired for jobs that were previously explicitly aimed at men, and vice versa.
- The share of women in companies that previously often specifically targeted men is increasing.

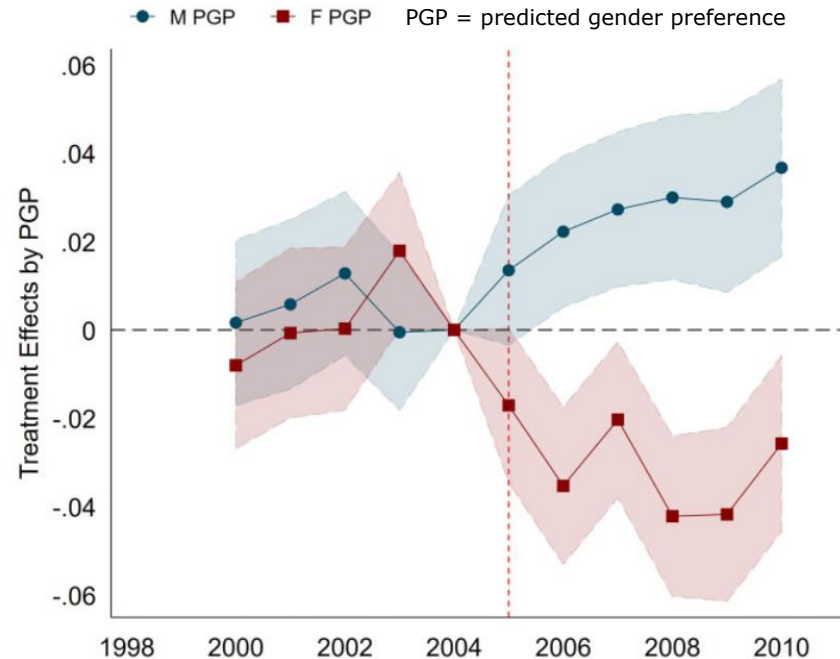


FIGURE 4

Probability that hired worker is female: event study results

Notes: This figure reports the estimation results for models of the event that a newly hired worker is female. Coefficients of the interaction term between year and indicators for vacancies classified as male or female PGP are reported. Solid lines represent point estimates, shadow areas show the 95% confidence intervals. Controls include firm, occupation and year fixed effects. The sample includes vacancies posted by frequent posting firms that posted at least two vacancies in the corresponding occupation in the pre-campaign period and in the post-campaign period (Column 6 of Table 1).

- Economic gender hierarchies are widely found in the labor market...
- Participation, salary, working hours, leadership level...
- These hierarchies are a global phenomenon; they do not change overnight.
- Several factors contribute to gender differences.
- → All factors we have discussed are closely interconnected!

Labor Market

Child Penalties

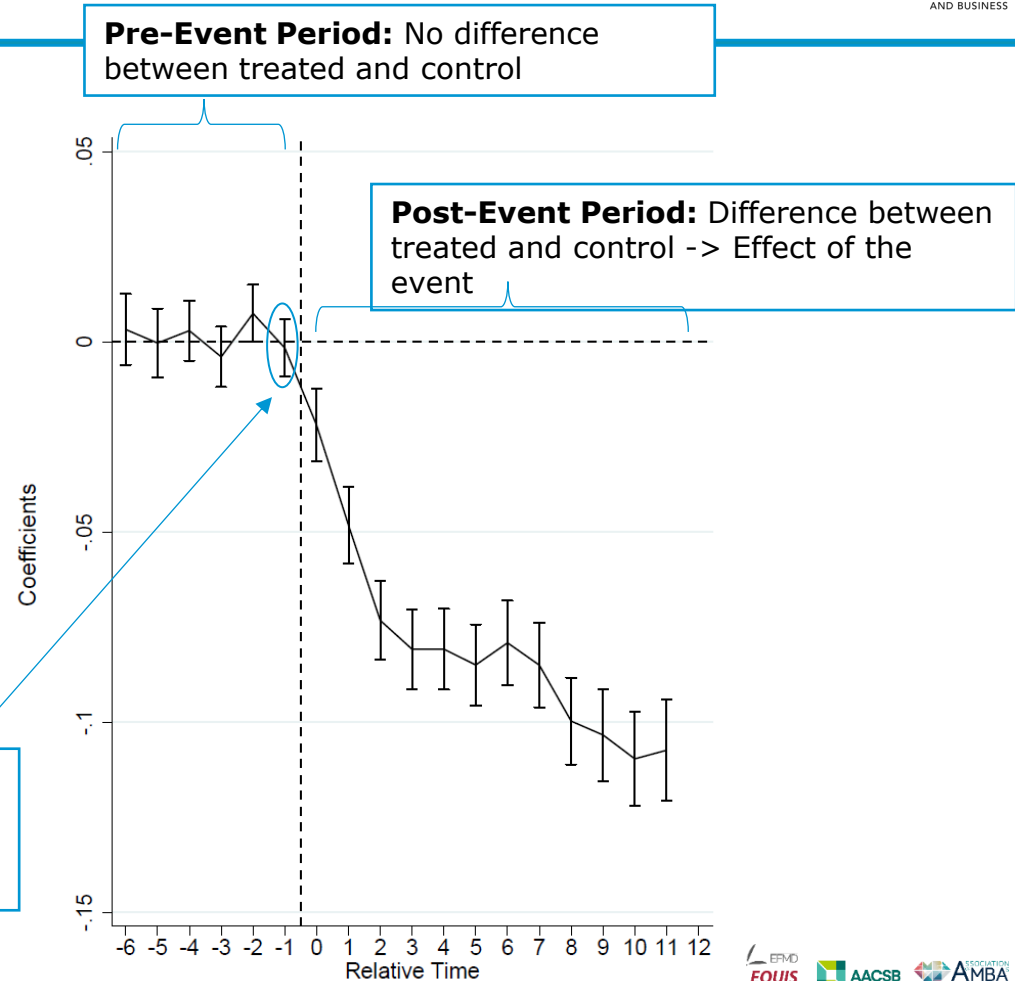


- **Kleven et al. (2019) Child Penalties across Countries: Evidence and Explanations**
- **Task:**
- read the paper
- Answer the question assigned to you for yourselves
- Find a person who has the other question -> discuss both questions
- Hold on to your thoughts & then share them in the large group
- **Question 1:** What research question does the paper pursue and why is it relevant? Explain the intuition of the estimation method!
- **Question 2:** Describe the main results of the paper. What do Figures 1-4 show? Did you expect the effects? What could be done to counteract them?

- **Kleven et al. (2019):**
- **Research Question:**
 - What effect does the birth of the first child have on the income of the parents?
 - Being a parent = crucial for the persistence of labor market inequalities
 - International differences (parental leave regulations, norms)
- **Method: Event-Study Estimation**
 - estimate the effect of the first birth on the income of the mother and father over a period of 10 years
 - Outcome: Income in year t relative to income directly before birth in $t = -1$
 - Intuition: In year t , a woman/man earns $X\%$ of the income in year $t = -1$
 - Calculation of the Child Penalty = Difference of the effect of the birth on mother & father/partner

Event Study Method

- Quasi-experimental method
- Comparison of the trends of the treatment & control group before/after an event
- Dynamics over time to estimate parallel trends
- Always displayed relative to the last time point before the change

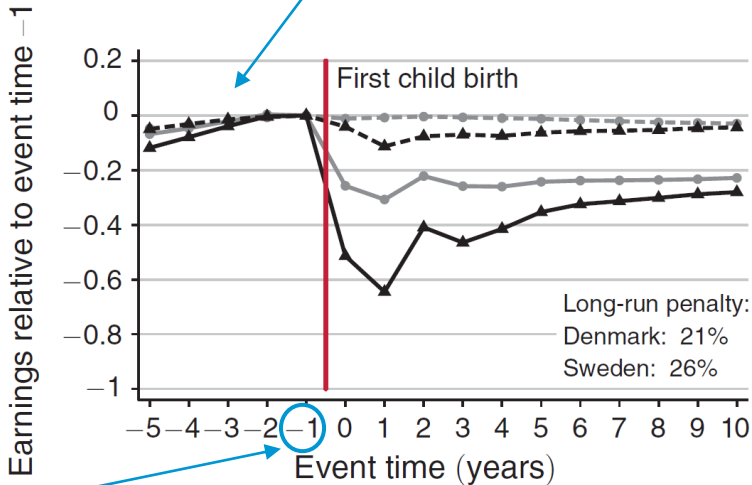


Labor Market

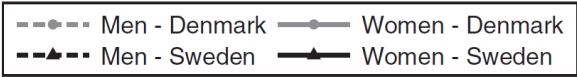
Child Penalties

Income for women declines after birth – also long-term!

Income development similar before birth



Income in the year before birth as a reference



CHILD PENALTIES IN EARNINGS IN SCANDINAVIAN COUNTRIES

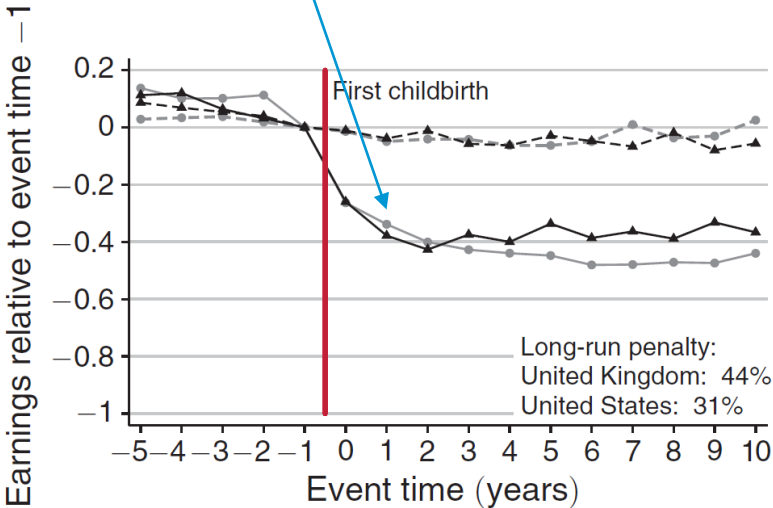


FIGURE 2. CHILD PENALTIES IN EARNINGS IN ENGLISH-SPEAKING COUNTRIES

Labor Market

Child Penalties

Birth has no effect on income for men – short- and long-term!

Why doesn't income increase over time? → Life-cycle and time trends are controlled for!

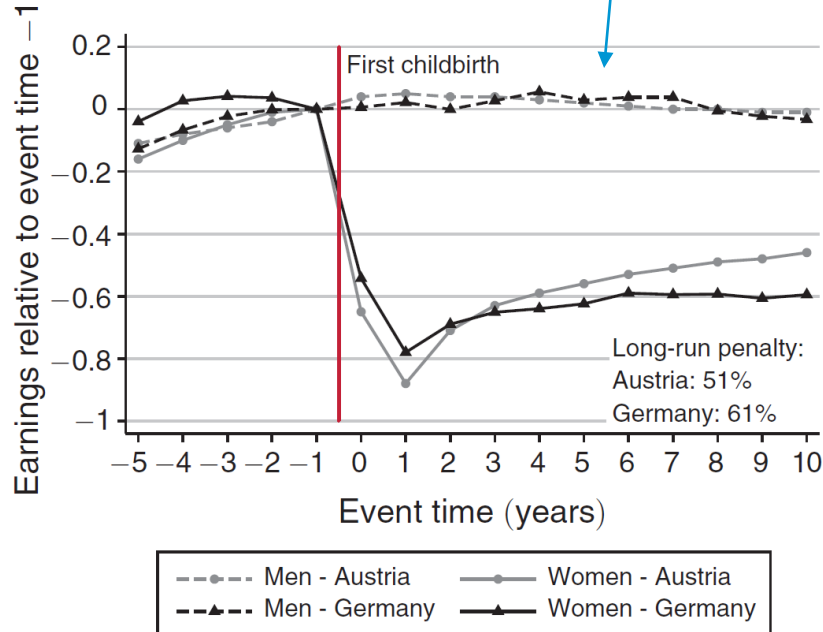


FIGURE 3. CHILD PENALTIES IN EARNINGS IN GERMAN-SPEAKING COUNTRIES

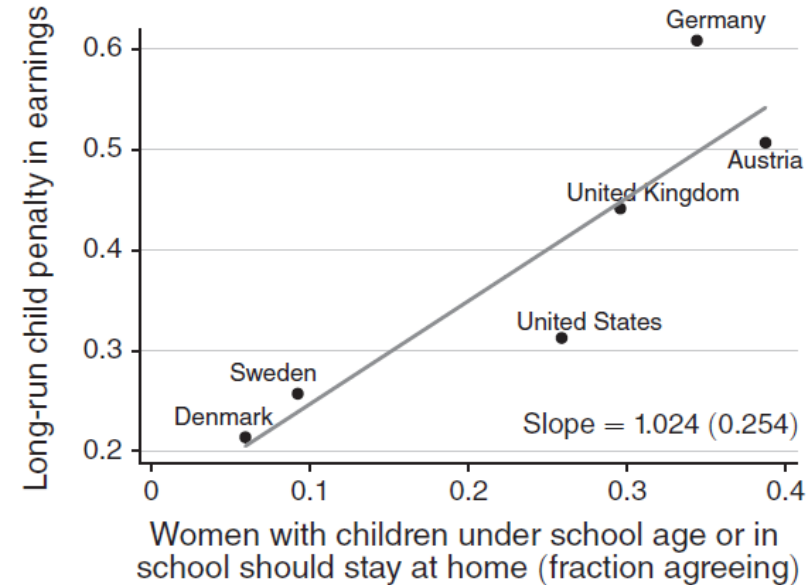


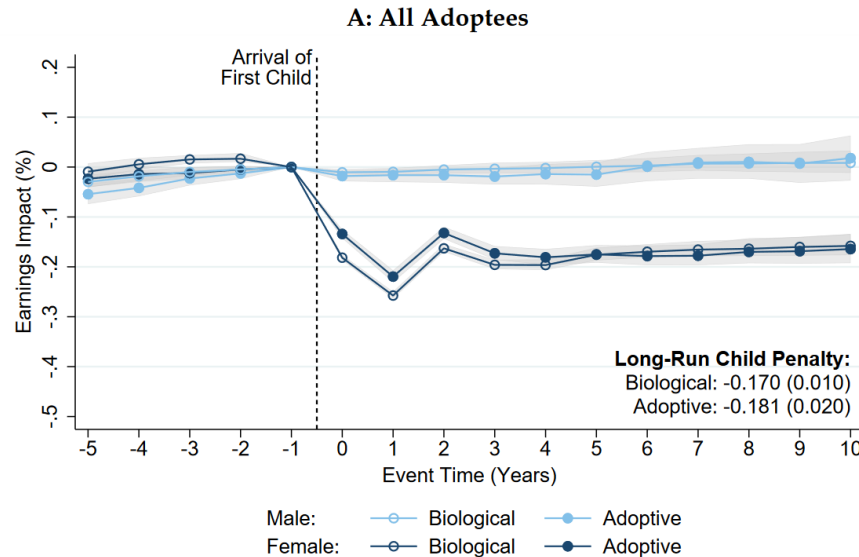
FIGURE 4. ESTIMATED CHILD PENALTIES VERSUS ELICITED GENDER NORMS

Does Biology Drive Child Penalties? Evidence from Biological and Adoptive Families

Kleven, Landais, Sogaard, *American Economic Review: Insights*, 2021

- Child Penalty also for adoptive parents
- Setting: Denmark

Figure 1: Child Penalties in Biological vs Adoptive Families



What causes the child penalty? Evidence from adopting and same-sex couples

ME Andresen, E Nix - Journal of labor economics, 2022

- Heterosexual and adoptive couples have very similar and persistent child penalties
- Strong drop in the mother's income
- Slight reaction in the father's income
- Child penalties for same-sex female couples are much lower and decrease over time
- The income of both mothers decreases, which indicates a very different approach to childcare
- Costs of birth are not a driving factor for child penalty
- Labor market advantages of fathers are not a driving factor for child penalty
- **Driving factors: Gender norms, preferences and discrimination against mothers in the labor market**

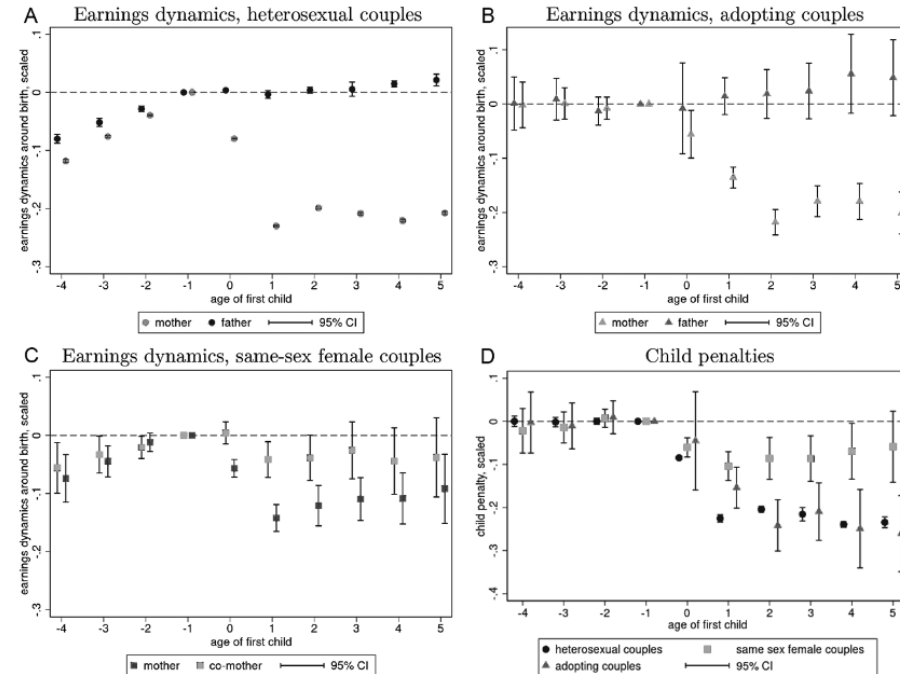


FIG. 1.—Estimated earnings dynamics (A–C) and child penalties (D). A–C show the earnings dynamics ($\alpha_{jc} + \beta_{jc}$ for mothers, β_{jc} for fathers) from equation (10), scaled with S_{jcm} from equation (11). D shows estimates of couple type–specific child penalties α_{jc} from equation (10), scaled with S_{jc1} . Sample construction and data are defined in section IV. Bootstrapped 95% confidence intervals (CIs) shown were determined using 200 replications and clustering by couple. Unscaled estimates can be found in figure C2. A color version of this figure is available online.

Culture, children and couple gender inequality

Jonas Jessen, European Economic Review 2022

- **Child Penalty East vs. West Germany**
- Comparison: Couples who grew up in the GDR/FRG and have children in unified Germany
- → Influence of socialization

Former West Germany:
Women's share of income declines more
Working hours decline more
With birth, care work increases

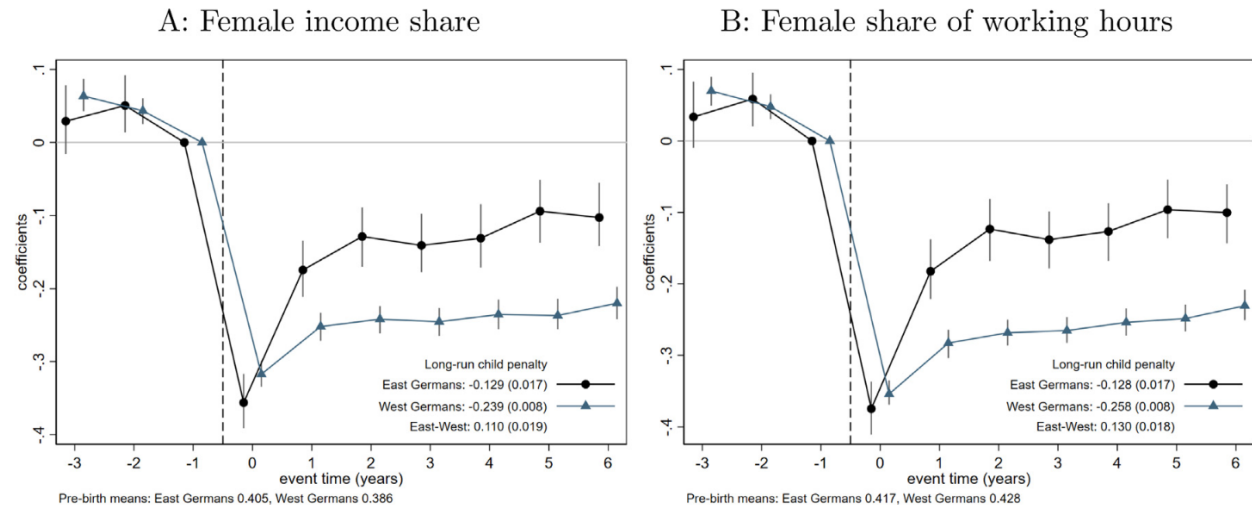


Fig. 2. Impact of children on labour market gender inequality. *Notes:* Figure shows event study estimates based on Eq. (2). Units of observation are couples. Coefficients are normalised to the pre-birth year ($t = -1$), means from this year are displayed in the figure notes. Long-run effects are defined as the average of post-birth ($t = 1 - 6$) coefficients with bootstrapped standard errors (500 replications). Income share refers to gross monthly labour income. East and West Germans are defined by their 1989 location.

Source: SOEP v36.

More on Child Penalties

- <https://childpenaltyatlas.org/>
- [Do Family Policies Shape the Evolution of Gender Inequality? Evidence from 60 Years of Policy Experimentation](#), forthcoming, *American Economic Journal: Economic Policy*, H. Kleven, C. Landais, J. Posch, A. Steinhauer and J. Zweimüller
- [The Geography of Child Penalties and Gender Norms: Evidence from the United States](#)
Kleven, NBER Working Paper #30176, January 2023